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Modelling the Efficiency of a Sheep Pasture in *Minecraft*

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Introduction

Minecraft is an open-world sandbox video game*. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep. Sheep can be sheared to gather wool, and must eat grass to regrow their wool after shearing. Grass blocks are converted to dirt after being eaten, and may be converted back to grass by other adjacent grass blocks. If a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. However, if the pasture is not densely packed, wool production will still be low because there are not many sheep. Between these two extremes, there is an optimal sheep population that will maximize wool output.

Deterministic model

Minecraft operates largely on chance. Both the growth of grass and the appetite of sheep is determined by random number generation. Every time the game's physics engine performs an update, grass has a chance to spread and sheep have a chance to eat. The grass population of the pasture is therefore governed by stochastic processes. Modelling stochastic processes is no simple task, so we will make a first attempt at modelling by assuming the pasture evolves according to deterministic, continuous-time processes.

Grass mechanics The *Minecraft* physics engine operates in discrete increments of time, called *ticks*. Every tick, a sheep has a chance to attempt to eat the block it's standing on. If it's standing on grass, it will successfully eat the grass and convert it into dirt. If the sheep and grass in the pasture are uniformly distributed, the expected number of sheep standing on grass is the product of the number of sheep and the fraction of grass blocks in the pasture. The rate at which grass is eaten is therefore given by:

$$\mu = \frac{NRN_g}{A} \tag{1}$$

Where μ is the rate at which grass is eaten, A is the size of the pasture, N_g is the number of grass blocks in the pasture, N is the number of sheep, and R is the frequency with which an individual sheep attempts to eat.

The rate at which grass grows not as simple. If N_g is small, then grass growth will be limited by the lack of places the grass can spread *from*. If N_g is large, grass growth will be limited by a lack of places for the grass to spread *to*. To capture this behavior, we will employ a logistic model, where population follows the differential equation:

*This investigation was conducted in Minecraft version 1.19.4.

$$\dot{y} = ky\left(1 - \frac{y}{p}\right) \tag{2}$$

Where $y(t)$ is the population at time t, k is a parameter representing the reproduction rate of the population and p is the carrying capacity, the maximum population that the local environment can sustain.

The logistic model is one of the oldest models for population growth. It was born from the older exponential (or *Malthusian*) model, but modified to include a maximum population size. It is useful as a tool of instruction, but has since been succeeded by more sophisticated and comprehentive models. However, we are not dealing with real-world ecology, but a video game which has simple population mechanics. Therefore the model is appropriate for our purposes. Let's define λ as the grass growth rate. Our population of interest is the number of grass blocks in the pasture N_g, and the carrying capacity is A:

$$\lambda = kN_g\left(1 - \frac{N_g}{A}\right) \tag{3}$$

Where k is a constant as described above.

The rate of population change is the difference of λ and μ:

$$\begin{aligned}\dot{N}_g = \lambda - \mu = kN_g\left(1 - \frac{N_g}{A}\right) - \frac{NRN_g}{A} \\ = \left(k - \frac{NR}{A}\right)N_g\left(1 - \frac{kN_g}{Ak - NR}\right)\end{aligned}$$

Equation (4) defines a logistic model for grass population with the effect of sheep:

$$\dot{N}_g = k^*N_g\left(1 - \frac{N_g}{A^*}\right) \tag{4}$$

Where

$$\begin{aligned}k^* = k - \frac{NR}{A} \\ A^* = A - \frac{NR}{k}\end{aligned}$$

N_g will asymptotically approach A^* as time goes on, therefore, we will refer to A^* as the equilibrium grass population.

Decay time A pasture in *Minecraft* will typically begin covered with grass, i.e. $N_g = A$. In the deterministic model, we will assume that the pasture has equilibrated and $N_g = A^*$, but the nonequilibium mechanics will become important for the stochastic model. Therefore we will determine the decay time here.

It would be very convenient if this asymptotic approach was exponential, as we could simply determine a characteristic decay time such as a half life. However, the approach is not exponential. The exact solution to (2) is:

$$y = \frac{p}{1 + \left(\frac{p-y_0}{y_0}\right) e^{-kt}} \tag{5}$$

Where y_0 is the initial population.

Since this is not an exponential decay, a characteristic decay constant cannot be defined. For comparative purposes, we'll proceed anyway and find the time, t_{90}, for which the asymptotic approach is 90% complete, i.e. $y(t_{90}) = y_0 - 0.9(y_0 - p)$:

$$t_{90} = -\frac{1}{k} \ln \left[\left(\frac{p}{y_0 - 0.9(y_0 - p)} - 1 \right) \left(\frac{y_0}{p - y_0} \right) \right]$$

For our logistic model in (4), we have a carrying capacity A^*, a rate constant k^*, and an intial population of A:

$$\begin{aligned} t_{90} &= -\frac{1}{k^*} \ln \left[\left(\frac{A^*}{A - 0.9(A - A^*)} - 1 \right) \left(\frac{A}{A^* - A} \right) \right] \\ &= -\frac{1}{k^*} \ln \left[\left(\frac{A - 0.9\frac{NR}{k} - 0.1\frac{NR}{k}}{A - 0.9(\frac{NR}{k})} - 1 \right) \left(\frac{A}{A^* - A} \right) \right] \\ &= -\frac{1}{k^*} \ln \left[\left(-\frac{0.1\frac{NR}{k}}{A - 0.9\frac{NR}{k}} \right) \left(\frac{-Ak}{NR} \right) \right] \\ &= -\frac{1}{k - \frac{NR}{A}} \ln \left[\frac{0.1A}{A - 0.9\frac{NR}{k}} \right] \\ &= \frac{1}{k - \frac{NR}{A}} \ln \left[\frac{A - 0.9\frac{NR}{k}}{0.1A} \right] \\ &= \frac{1}{k - \frac{NR}{A}} \ln \left[10 - 9\frac{NR}{Ak} \right] \end{aligned} \tag{6}$$

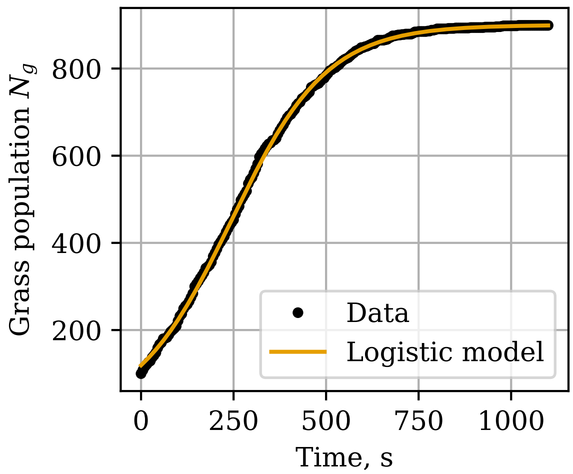
Experimental determination of rate constants Before proceeding further, it's worth determining the actual values of k and R. This will be done via in-game experiments, which will be described briefly here. More detail can be found on the author's github†.

Ten sheep were locked in cells with a grass block immediately underneath them. The cells included circuitry to count when each sheep ate the grass and automatically replenish it. In one hour, the sheep consumed 320 blocks of grass. This gives an R value of 0.00888 s^{-1}.

A 30x30 patch of dirt was isolated from its surroundings and uniformly interspersed with 100 grass blocks. As the grass grew, the grass population was recorded every 5 seconds. The data was fit to the exact solution of the logistic equation (5). The best fit gave $k = 0.00773$ s^{-1}. The data and fit are compared in Figure 1. The close fit gives us also confidence that the logistic model is appropriate.

†github.com/ELFREELAND/minecraft_sheep

Figure 1 Logistic model fit to grass growth data.



The fast farmer For simplicity, let's imagine the farmer has superhuman speed, and is able to shear each sheep the instant it grows its wool. In this case, the rate at which wool is gathered is equal to μ. Denoting the rate of wool gathering with F, and substituting $N_g = A^*$ into (1):

$$F_{fast} = \mu = NR - \frac{N^2R^2}{Ak} \tag{7}$$

We can find the optimal sheep population by differentiating (7) w.r.t. N, setting the derivative to 0 and solving for N:

$$N = \frac{Ak}{2R} \tag{8}$$

This is the sheep population which will produce the maximum wool output, if the farmer is significantly fast enough. At the optimal value of N, the rate of wool production is:

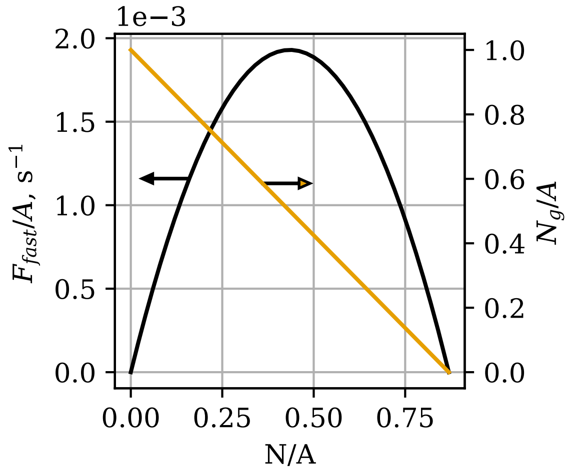
$$F_{fast} = \frac{Ak}{4}$$

Figure 2 shows F_{fast} and N_g as a function of N, normalized for pasture area. These are simple curves, a parabola and straight line respectively, but there are a few important features to glean. First, the equilibrium N_g and F_{fast} are both zero at $\frac{N}{A} = \frac{k}{R} = 0.869$. Deterministically, the grass in a pasture with $N > 0.869A$ will become extinct.

Second, the maximum F_{fast} coincides with $N_g = 0.5A$. This is a feature of the logistic model; population growth is greatest when the population is half of the carrying capacity.

The slow farmer *Minecraft* players typically don't want to put all of their time into shearing their sheep as quickly as they can. For a more realistic model, we'll assume the farmer shears all sheep at once, intermittently and at regular intervals.

Figure 2 Plot of $\frac{F_{fast}}{A}$ and $\frac{N_g}{A}$ as a function of N, normalized to pasture size.



The farmer is off doing other things, and every T seconds, they come back to shear the sheep. Upon shearing the sheep, all the sheep are wool-less. By the time they come back for the next shearing, the sheep have regrown some amount of wool. Let's say the number of sheep that have their wool is N_w. So when the farmer shears the sheep, they get N_w wool. Then the average rate at which the farmer gathers wool is

$$F_{slow} = \frac{N_w}{T}$$

Now we need to consider the value of N_w. A sheep will only regrow wool upon eating if it doesn't already have any, so we should expect that the rate at which sheep are regrowing wool is the rate at which they are eating, μ, times the fraction of sheep that are wool-less:

$$\dot{N}_w = \frac{N - N_w}{N}\mu = (N - N_w)\frac{RN_g}{A} \tag{9}$$

At time $t = 0$, the farmer shears all the sheep, leaving them with $N_w = 0$, and then at time $t = T$, the farmer comes back and shears them again, gathering N_w blocks of wool. How many sheep grew their wool back in time T? We can find out by solving (9) for N_w. We'll separate it into N_w terms and non N_w terms:

$$\frac{1}{N - N_w} dN_w = \frac{RN_g}{A} dt$$

And integrate from initial state $(t = 0, N_w = 0)$ to the final state $(t = T, N_w = N_{w,T})$. We're assuming the pasture has reached steady state, so N_g is independent of t and can be treated as a constant.

$$\int_0^{N_{w,T}} \frac{1}{N - N_w} dw = \int_0^T \frac{RN_g}{A} dt$$

$$-\ln\left(\frac{N - N_{w,T}}{N}\right) = \frac{TRN_g}{A}$$

And solve for $N_{w,T}$:

$$N_{w,T} = N - Ne^{\frac{TRN_g}{A}}$$

So when the farmer comes back to shear the sheep at $t = T$, they will get $N - N\exp\left(-\frac{TRN_g}{A}\right)$ blocks of wool. If the farmer keeps coming back every T seconds, they will be gathering wool at a rate

$$F_{slow} = \frac{N - Ne^{-\frac{TRN_g}{A}}}{T}$$

Finally, we can plug (**??**) for N_g:

$$F_{slow} = \frac{N - Ne^{-TR+\frac{TR^2N}{Ak}}}{T} \tag{10}$$

This is the expression for the rate of wool gathering by a slow farmer. To find the optimal sheep population, we can take the same approach as we did with the fast farmer case. Differentiating (10) w.r.t. N and setting $F_{slow} = 0$:

$$\frac{dF_{slow}}{dN} = \frac{1 - \left(\frac{TR^2N}{Ak} + 1\right)e^{-TR+\frac{TR^2N}{Ak}}}{T} = 0$$

Solving for N, we obtain the optimal value of N:

$$N = \frac{Ak}{TR^2}(W(e^{TR+1}) - 1) \tag{11}$$

Where W is the Lambert W function, which satisfies $W(xe^x) = x$. Plugging this back into (10), we obtain the value of F_{slow} at optimal N:

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR+W(e^{TR+1})-1})}{R^2T^2} \tag{12}$$

Figure 3 shows a plot of (10), and figure 4 shows plots of (11) and (12). As T increases, the optimal sheep population also increases, and the optimal rate of wool gathering decreases. Clearly there is a tradeoff between the frequency with which the farmer shears the flock and the rate at which they gather wool.

There are a couple important things to note about F_{slow}. First, compare the plot of F_{fast} in figure 2 with the plot of F_{slow} with $T = 1$ in figure 3b - they are almost identical. We can see this algebraically by employing the following approximation, for $x \ll 1$:

$$e^x \approx x + 1$$

In equation (10), the exponent is small for small values of T, therefore:

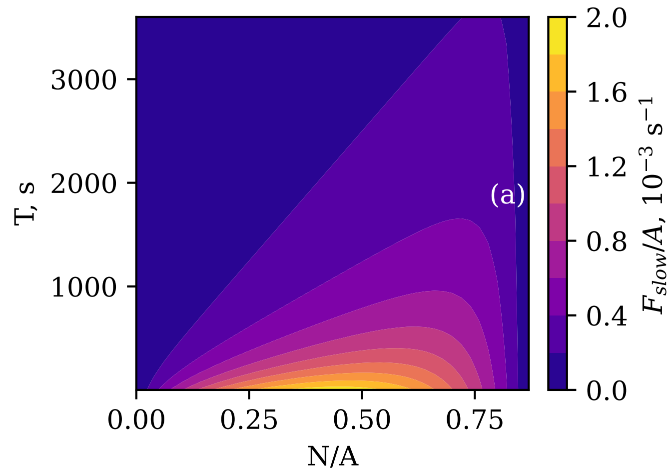
$$F_{slow} \approx \frac{N - N(-TR + \frac{TR^2N}{Ak} + 1)}{T} = NR - \frac{N^2R^2}{Ak} = F_{fast}$$

Second, it appears in figure 3b that for large values of T and modest values of N, F_{slow} is approximately linear w.r.t. N. Under these conditions, the exponent $-TR + \frac{TR^2N}{Ak}$ in (10) is large and negative, and:

$$F_{slow} \approx \frac{N - 0}{T} = \frac{N}{T}$$

Physically, this represents conditions for which all of the sheep have regrown their wool by the time the farmer comes back to harvest, so the farmer is simply gathering N blocks of wool with every harvest. Lines of $\frac{N}{T}$ are plotted over top of the curves in figure 3a.

Figure 3 (a) Heatmap plot of F_{slow} as a function of T and N. (b) Contours of F_{slow} as a function of N at constant T.



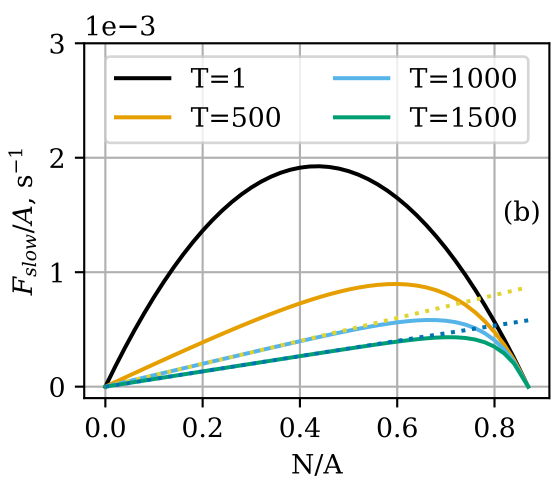
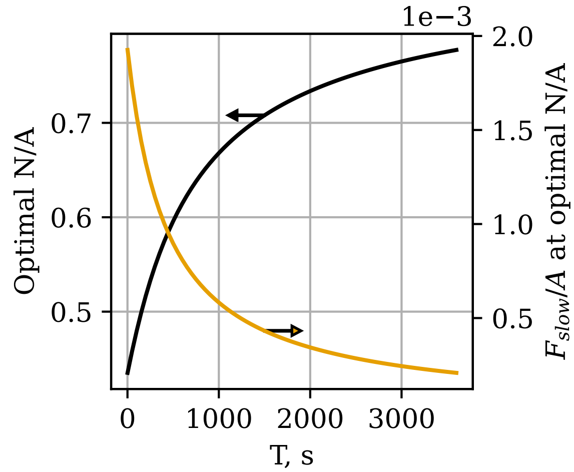


Figure 4 Optimal sheep population and maximum wool production as a function of T, normalized to pasture size.



Stochastic model

The deterministic model gives valuable insight into the expected behavior of the pasture, but a more complete picture can be obtained by developing a stochastic model which captures the randomness inherent in the physics engine of *Minecraft*. We will develop a stochastic model for the grass population in the pasture using Kolmogorov equations. This approach for population modelling is described in more detail in [**1**], but the key elements will be outlined here.

Grass Mechanics In the deterministic model, we considered the grass population as a single number, N_g, which evolved with time. In the stochastic approach, we will consider the evolution of the *probability distribution* of N_g with time.

We'll introduce the following notation:

$$p_x(t) = \Pr(N_g(t) = x)$$

In words, $p_x(t)$ is the probability that the pasture has $N_g = x$ at time t.

We'll also need to slightly reconsider the meanings of μ and λ. In the deterministic model, these are the rate of grass growth or consumption. For a small time interval Δt, the number of grass blocks that are eaten is $\mu\Delta t$, and likewise for λ. If Δt is sufficiently short so that the probability of two independent changes in grass population is negligible, $\mu\Delta t$ can be reinterpreted as the probability of one grass block being consumed in time Δt, and likewise for λ. μ and λ therefore represent the probability of the pasture transitioning to a state with one more or one less grass block than before.

Using these new interpretations of μ and λ, we can write a system of differential equations describing the evolution of the probability distribubion of N_g with time:

$$\dot{p}_x(t) = -p_x(t)(\lambda(x) + \mu(x)) + p_{x-1}(t)\lambda(x-1) + p_{x+1}(t)\mu(x+1) \qquad (13)$$

Note that $\mu(x)$ simply represnts the value of μ with $N_g = x$, and likewise for λ.

Systems of equations such as these, which describe the evolution of a probability distribution with time, are known as *Kolmogorov equations*. We will have a system of $A+1$ equations, one for each possible value of N_g.

There are a couple of special cases to address, namely, the cases of $\dot{p}_0$ and $\dot{p}_A$. Since $\lambda(0) = \mu(0) = 0$ and $p_{-1} = p_{A+1} = 0$:

$$\dot{p}_0(t) = p_1(t)\mu(1)$$

$$\dot{p}_A(t) = -p_A(t)(\lambda(A) + \mu(A)) + p_{A-1}(t)\lambda(A-1)$$

We'll wrap up the system into vector form for convenient expression. Let $\mathbf{p}$ be a vector of p_x, and $\dot{\mathbf{p}}$ its derivative:

$$\mathbf{p}(t) = [p_0(t), p_1(t), p_2(t) \ldots p_A(t)]$$

$$\dot{\mathbf{p}}(t) = \frac{d\mathbf{p}}{dt} = [\dot{p}_0(t), \dot{p}_1(t), \dot{p}_2(t) \ldots \dot{p}_A(t)]$$

Let $\mathbf{R}$ be a matrix of coefficients relating $\mathbf{p}$ and $\dot{\mathbf{p}}$:

$$\mathbf{R} = \begin{bmatrix} r_{0,0} & r_{0,1} & r_{0,2} & \cdots & r_{0,A} \\ r_{1,0} & r_{1,1} & r_{1,2} & \cdots & r_{1,A} \\ r_{2,0} & r_{2,1} & r_{2,2} & \cdots & r_{2,A} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{A,0} & r_{A,1} & r_{A,2} & \cdots & r_{A,A} \end{bmatrix}$$

$$r_{i,j} = \begin{cases} r_{0,j} = 0 \\ r_{i,i+1} = \lambda(i) \\ r_{i+1,i} = \mu(i) \\ r_{i,i} = -(\lambda(i) + \mu(i)) \\ 0 \ \ \text{otherwise} \end{cases}$$

Then:

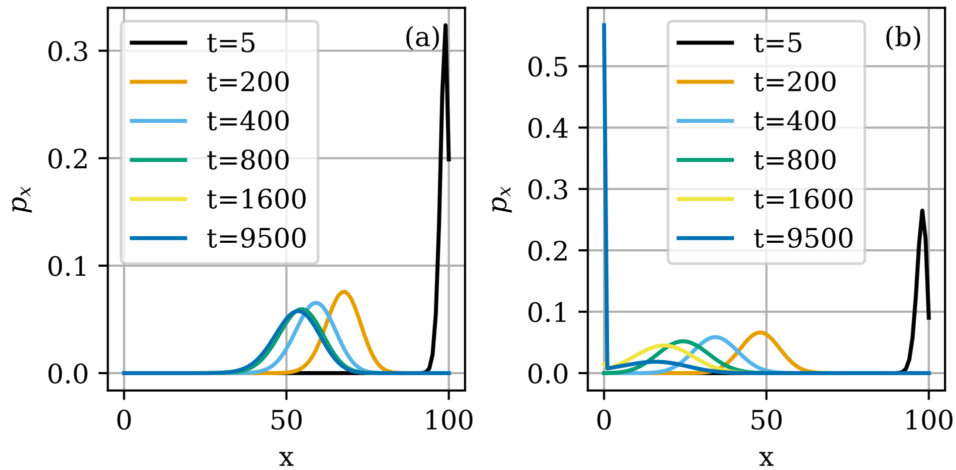
$$\dot{\mathbf{p}}(t) = \mathbf{p}(t)\mathbf{R}$$

All that's left to do now is to pick values for N and A and an initial value for N_g and solve. This is most easily done numerically, as has been done here using the Runge-Kutta method.

Figure 5 shows the evolution of the distribution of N_g for $N = 40$ and $N = 70$ in a 100-block pasture. Both pastures begin with $N_g = 1$, i.e. $\mathbf{p} = [0, 0, 0, \ldots 0, 1]$. The pasture with $N = 40$ approaches a stable state, evidenced by the indistinguishable distributions at $t = 1600$ s and $t = 9500$ s. In contrast, the pasture with $N = 70$ continues evolving toward smaller and smaller grass populations, with p_0 increasing dramatically as time proceeds.

We can already see the stochastic model deviating from the deterministic model. A pasture with $N/A = 0.7$ equilibrates at $N_g = 0.196A$ in the deterministic model, but the stochastic model is predicting a high likelihood of complete extinction on the time scale of a few hours.

Figure 5 Evolution of p_x distribution for a pasture of size $A = 100$, for (a) $N = 40$ and (b) $N = 70$.



The reason for this isn't hard to see by considering the equation for $\dot{p}_0$:

$$\dot{p}_0(t) = p_1(t)\mu(1)$$

Since p_1 and $\mu(1)$ are always positive, p_0 is monotonically increasing, bounded above by $p_0 = 1$. Therefore the true equilibrium distribution is $\mathbf{p} = [1, 0, 0, 0, \ldots]$ i.e. the grass has become exinct. However, even if this is the equilibrium state, the pasture may reach a different *metastable* state if the probability of extinction remains negligible across reasonable time scales. This is what is seen in figure 5a. The distribution for $N = 40$ "stabilizes" with a value of p_1 so low that $\dot{p}_0$ remains negligible.

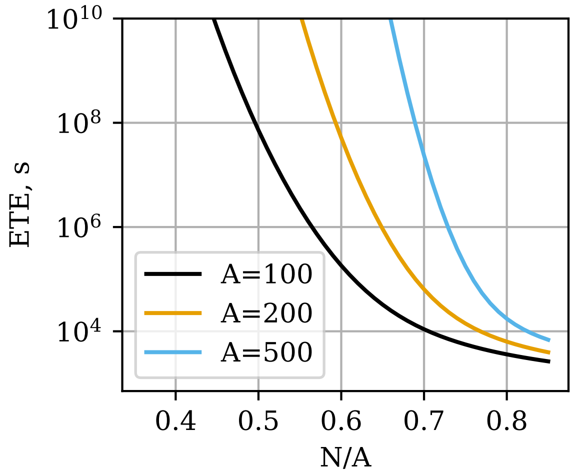
This raises two obvious questions: First, what is the expected time to extinction, i.e. how high can N be without causing a significant increase in p_0 over reasonable time scales? Second, for pastures where $\dot{p}_0$ is negligible, what is the metastable distribution of N_g? We'll examine grass extinction first.

Grass extinction Rapid grass extinction at high sheep densities poses a problem, since the slow farmer deterministic model predicts that the optimal N increases as T increases (see figure 4). Fortunately, an exact solution can be obtained for the expected time to extinction ETE. Let $\mathbf{M}$ be a matrix of mean residence times, such that the element m_{ij} of $\mathbf{M}$ is the expected amount of time the pasture spends with $N_g = j$ before reaching extinction, given that it began with a population i. It can be shown [**3**] that $\mathbf{M}$ is obtained by removing the first row and column from $\mathbf{R}$ and taking the negative of the inverse of this modified $\mathbf{R}$ matrix. Denoting the modified $\mathbf{R}$ matrix $\mathbf{R}^\dagger$:

$$\mathbf{M} = -(\mathbf{R}^\dagger)^{-1}$$

The sum of the ith row of $\mathbf{M}$ gives the total time spent with nonzero population before becoming extinct, given the pasture started with $N_g = j$. The typical *Minecraft* pasture begins fully populated with grass, i.e. $N_g(0) = A$. Therefore the value we are interested in is:

Figure 6 Expected time to extinction for pastures of $A = 100$, 200, and 500 as a function of sheep population density.



$$\text{ETE} = \sum_j m_{Aj}$$

We can apply this solution to obtain the ETE for any value of N and A[‡], as demonstrated in figure 6. For populations close to the deterministic sheep population limit $\frac{N}{A} = \frac{k}{R} = 0.869$, the ETE is on the order of a few hours for all three pasture sizes. As the sheep population density decreases, the ETE increases dramatically - For a 100 block pasture with a sheep density of 0.5, the ETE is on the order of several years. ETE also increases dramatically with larger pastures - A 100-block pasture with sheep density 0.7 has ETE on the order of a few hours, while a 500-block pasture with the same density has ETE on the order of months.

Metastable distribution Let's now turn our attention to the metastable, or quasi-equilibrium, state of the pasture. The pasture will cease evolving when all $\dot{p}_x = 0$. This implies that the likelihood of the pasture having $N_g = x$ and losing one grass block equals the probability of $N_g = x - 1$ and growing one grass block [**4**]. In symbols:

$$p_x \mu(x) = p_{x-1} \lambda(x-1) \tag{14}$$

We will assume $N_g = 0$ is an unreachable state and exclude it from consideration. Writing (14) for $2 \leq x \leq A$, and combining with the normalization condition that $\sum_{x=1}^{A} p_x = 1$, we obtain a system of A equations in A variables:

[‡]Although, floating point arithmetic may become a problem for small population densities as the numbers involved become extremely large.

$$\begin{cases} p_2\mu(2) - p_1\lambda(1) = 0 \\ p_3\mu(3) - p_2\lambda(2) = 0 \\ \dots \\ p_A\mu(A) = p_{A-1}\lambda(A-1) = 0 \\ \sum_{x=1}^{A} p_x = 1 \end{cases} \tag{15}$$

We can solve (15) to obtain the metastable N_g distribution. Figure 7 shows the metastable distribution for a few select conditions, and figure 8 shows the expected value and standard deviation of the metastable distribution as a function of sheep density. There are a few important features to note. First, figure 7 shows both the metastable distribution predicted by (15) and numerical solutions of the Kolmogorov equations for $t = 10000$. The results match in each case except for the case $A = 100$, $N = 65$. In that case, the numerical solution has already begin to approach extinction. This is the only case in which the metastable p_1 is significant, and it is also the only case in which p_0 is significant in the numerical solution. Intuitively, one might gather that for conditions where the metastable p_1 is significant, the pasture will not remain in the metastable state for long as p_0 will rapidly increase.

This perspective can explain the variation of ETE with pasture size and sheep population. As N increases, the metastable distribution shifts to smaller values of N_g (figure 8a) and broadens (figure 8b), both of which will increase p_1 and thereby $\dot{p}_0$. This is why larger sheep densities decrease ETE (figure 6).

As pasture size increases at constant sheep population density, the relative width of the distribution decreases. What is meant by relative width here is the broadness of the distribution as a fraction of the width of its domain, which corresponds here to the standard deviation divided by A. Compare figure 7a and 7b, and observe how the distributions for $A = 500$ are visually narrower than those at $A = 100$ for the same $\frac{N}{A}$. Of two pastures with the same $\frac{N}{A}$, the one with larger A will have a narrower distribution, and therefore the expected N_g can move to smaller values before p_1 becomes significant. This is why larger pasture sizes increase ETE (figure 6).

Figure 8a shows that the expected metastable grass density is approximately the same as the equilibrium grass density from the deterministic model (compare figure 2). This trend is present up to at least $\frac{N}{A} = 0.5$ for the conditions chosen here, but already some deviation is visible for high sheep densities.

The slow farmer let

$$q_x = \Pr(N_w(t) = x)$$

then

$$\dot{q}_x(t, N_g) = q_{x-1}(t)\frac{N-(x-1)}{N}\mu(N_g) - q_x(t)\frac{N-x}{N}\mu(N_g)$$

$$= \frac{\mu(N_g)}{N}[q_{x-1}(t)(N-x+1) - q_x(t)(N-x)]$$

$$\dot{\mathbf{q}}(t) = \mathbf{q}(t)\mathbf{S}$$

Figure 7 Metastable N_g distributions for pastures with (a) $A = 100$ and (b) $A = 500$. Numerical solutions for p_x at $t = 10000$ are shown as dotted lines.

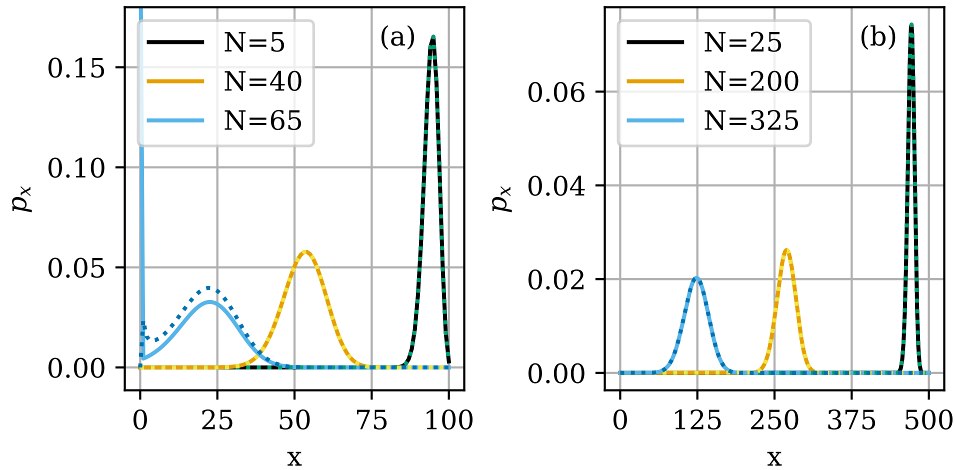
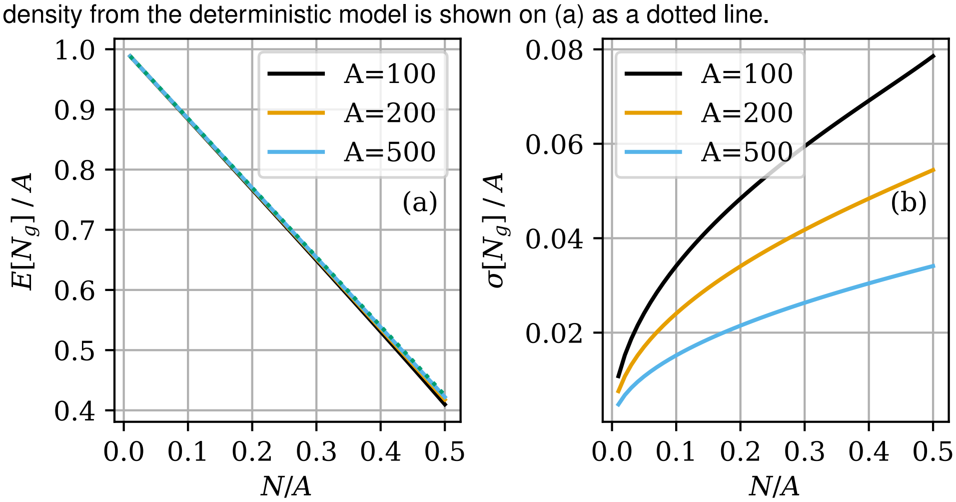


Figure 8 (a) Expected grass population density and (b) standard deviation normalized to pasture size, as a function of sheep population density. The equilibrium grass density from the deterministic model is shown on (a) as a dotted line.



$$\mathbf{S} = \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & \cdots & s_{0,N} \\ s_{1,0} & s_{1,1} & s_{1,2} & \cdots & s_{1,N} \\ s_{2,0} & s_{2,1} & s_{2,2} & \cdots & s_{2,N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{N,0} & s_{N,1} & s_{N,2} & \cdots & s_{N,N} \end{bmatrix}$$

$$s_{i,j} = \frac{\mu(N_g)}{N} * \begin{cases} s_{i,i+1} = N - i \\ s_{i,i} = N - i \\ 0 \ \text{ otherwise} \end{cases}$$

for some N_g, $\mathbf{q}(T)$ is the solution of the above for $\mathbf{q}(0) = [1, 0, 0, \ldots]$

the final actual $\mathbf{q}(T)$ is the sum of all possible $\mathbf{q}(T)$ for all N_g, weighted by the metastable p_x

Conclusion

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